

Digital Egypt Pioneers Initiative

Round 4

2025 -2026

AI-POWERED PREDICTIVE MAINTENANCE FOR INDUSTRIAL EQUIPMENT

Track: AI & Data Science - Microsoft Machine Learning

Engineer Group Code: ONL4_AIS2_S7

Instructor: Eman Elgalad

Company Name: Skills Dynamix

TEAM ROLES

Team Members	Role	Responsibilities
Moaz Osama Hataba (TL)	Project Manager & System Architect	Architecture design, team coordination, integration, documentation
Abrar Ashraf Shaban	Data Engineer & Machine Learning Engineer	Data preprocessing, feature engineering, Model development, XGBoost training, performance evaluation
Nour-Eldin Kamel Muhammad	XAI Engineer	SHAP explainability, model comparison
Salma Muhammad Nageh	Backend & Authentication Engineer	Backend services, authentication, business logic, system integration
Marwan Muhammad Darwish	Frontend & RAG Engineer	Streamlit dashboard, UI development, RAG assistant integration
Yussef Ahmad Muhammad	Database, Reporting	Database design, reports, & exports

PROJECT OVERVIEW

- Transform traditional maintenance from a reactive approach into a predictive and intelligent process using Artificial Intelligence.
- Develop an Explainable AI (XAI) system to predict the Remaining Useful Life (RUL) of aircraft engines.
- Utilize NASA CMAPSS dataset to analyze engine degradation through sensor measurements and operational conditions.
- Combine Machine Learning models with SHAP explainability to provide accurate and transparent predictions.
- Deliver an interactive Streamlit dashboard for real-time RUL prediction, health monitoring, and maintenance insights.

PROBLEM STATEMENT

- Limitations of Traditional Approaches

1. Reactive Maintenance:

Repairs are performed only after failure occurs, leading to unexpected downtime.

2. Preventive Maintenance:

Fixed maintenance schedules may result in unnecessary interventions or missed failures.

- Industrial equipment failures can lead to:

- Unexpected failures interrupt operations and reduce productivity.
- High Maintenance Costs due to emergency repairs and unnecessary servicing.
- Many AI models provide predictions without clear reasons, reducing trust in automated decisions.
- Safety Risks in critical industrial environments.

OBJECTIVES

Develop an Explainable AI-driven Predictive Maintenance System that predicts equipment failures before they occur by estimating Remaining Useful Life (RUL) and supporting proactive maintenance decisions.

- Analyze the NASA CMAPSS dataset and engine degradation behavior.
- Explore sensor patterns, trends, and operational conditions.
- Develop a machine learning model to estimate the remaining operational cycles of equipment before failure.
- Analyze sensor measurements and operational conditions to identify equipment degradation patterns.
- Provide Explainable AI Insights.
- Develop an Interactive Monitoring Dashboard.
- Enable proactive maintenance planning to reduce downtime, costs, and unexpected failures.

TECHNOLOGY STACK

AI & Machine Learning

Python, XGBoost, Scikit-learn, SHAP,
Pandas, NumPy, RAG, & ML Flow

Frontend

Streamlit, Plotly, & Matplotlib

Authentication & Security

bcrypt

Database

SQLite, & FAISS

Reporting

ReportLab, & OpenPyXL

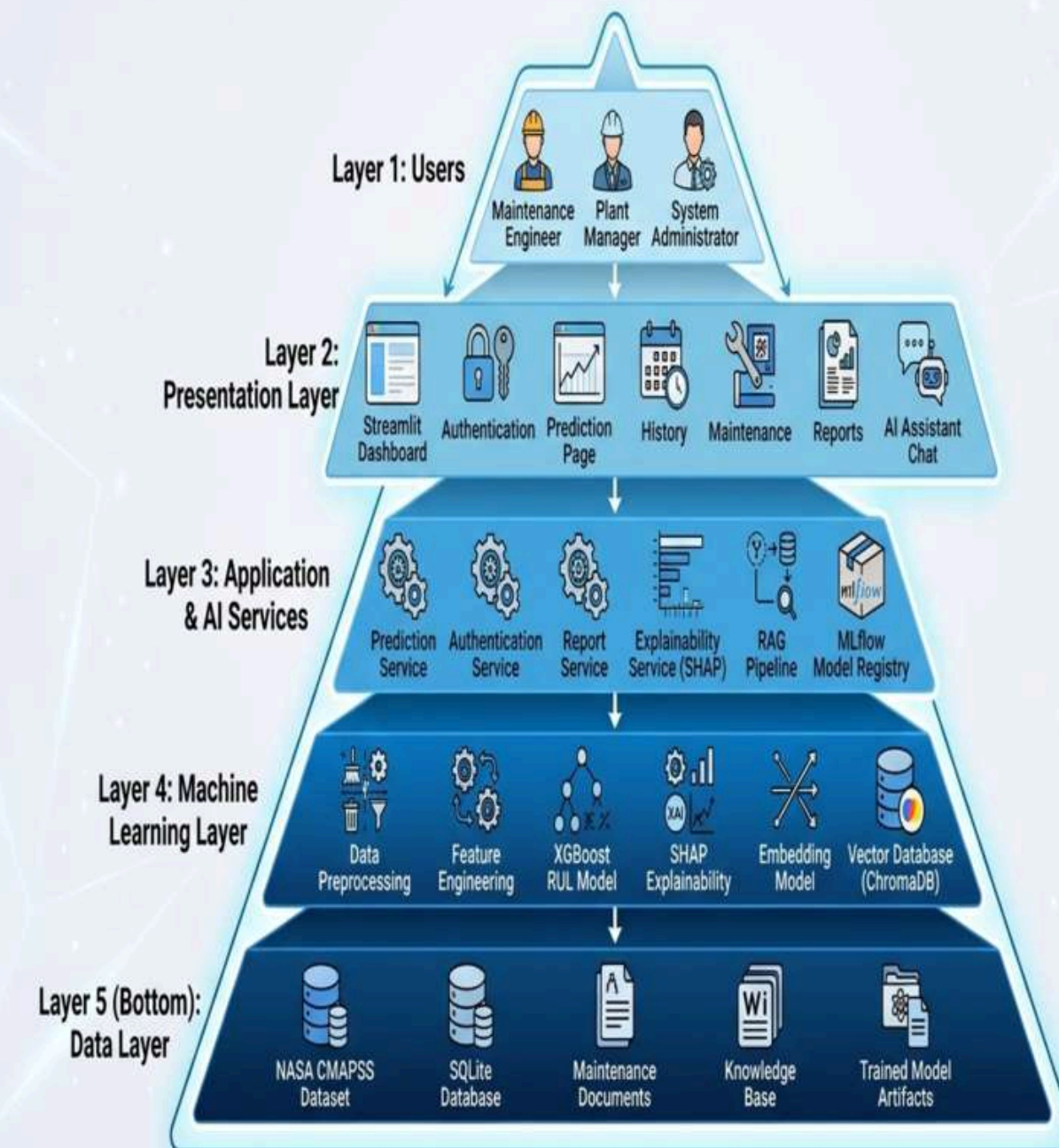
Development

GitHub, & VS Code

SYSTEM ARCHITECTURE

- The system follows a layered architecture to ensure modularity and maintainability. Users interact with the Streamlit dashboard, which communicates with the application services. These services invoke the AI engine for preprocessing, prediction, and explainability, while all application data is stored in the SQLite database. This separation of concerns makes the system easier to maintain, extend, and test.

AI-DRIVEN PREDICTIVE MAINTENANCE SYSTEM WITH XAI ARCHITECTURE



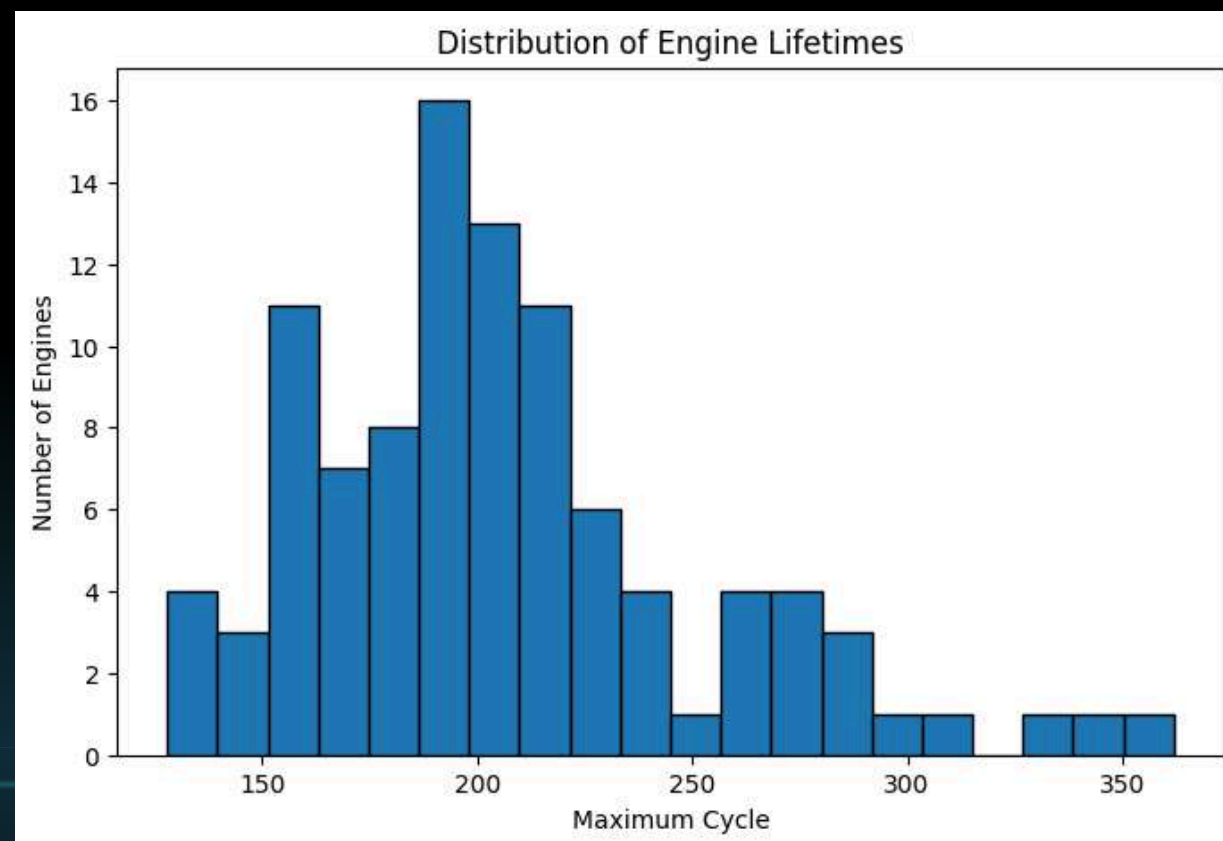
DATASET OVERVIEW

- **Source:** NASA Commercial Modular Aero-Propulsion System Simulation (CMAPSS)
- **Type:** Multivariate time-series dataset
- **Purpose:** Simulates the degradation of aircraft turbofan engines under different operating conditions.
- **Scenarios:** Four datasets (FD001–FD004) representing different fault modes and operating conditions.
- **Data Includes (26 features):**
 - Engine ID.
 - Operating Cycle.
 - 3 Operational Settings.
 - 21 Sensor Measurements.
- **Train:** 20,631 records
- **Test:** 13,096 records
- **100 Test Engines**
- **Target:** Remaining Useful Life (RUL)

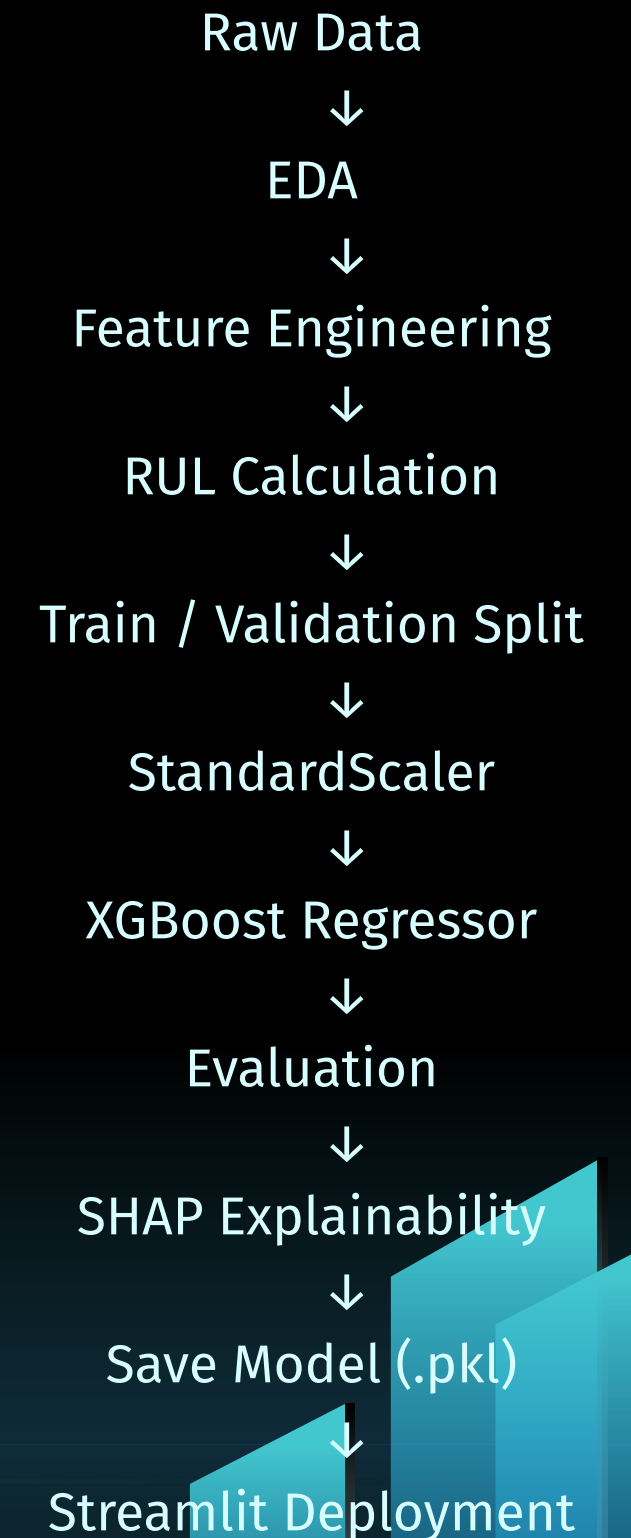
MACHINE LEARNING PIPELINE

Data Cleaning

- Removed constant features.
- Removed highly correlated feature (sensor_14).
- Calculated RUL.
- Applied RUL clipping (125 cycles).



	unit	cycle	RUL
0	1	1	125
1	1	2	125
2	1	3	125
3	1	4	125
4	1	5	125
5	1	6	125
6	1	7	125
7	1	8	125
8	1	9	125
9	1	10	125
10	1	11	125



TRAIN / VALIDATION / MODEL

Dataset	Engines
Train	80
Validation	20

Dataset	Samples
Train	16,561
Validation	4,070

XGBoost Regressor

- `n_estimators = 300` , `learning_rate = 0.05` , `max_depth = 6`

Validation Results

RMSE : 16.950
MAE : 12.157
 R^2 : 0.8349

Test Results

RMSE : 17.743
MAE : 13.085
 R^2 : 0.8177

The close validation and test performance indicates good generalization with no significant overfitting.

Split by engine IDs to avoid Data Leakage.

MODEL COMPARISON

Model	RMSE ↓	MAE ↓
XGBoost (Final Model)	16.95	12.16
LSTM	17.49	12.55

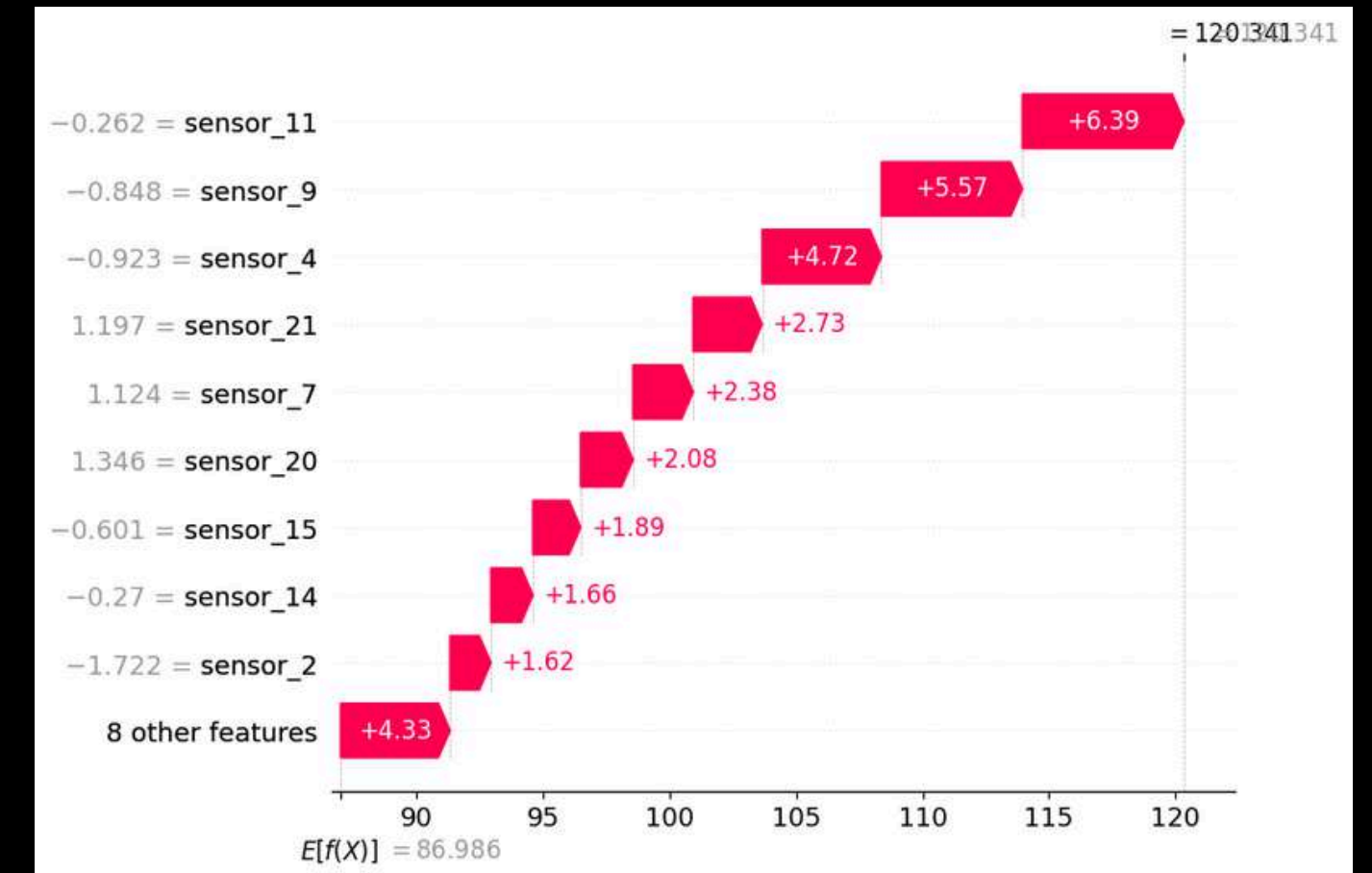
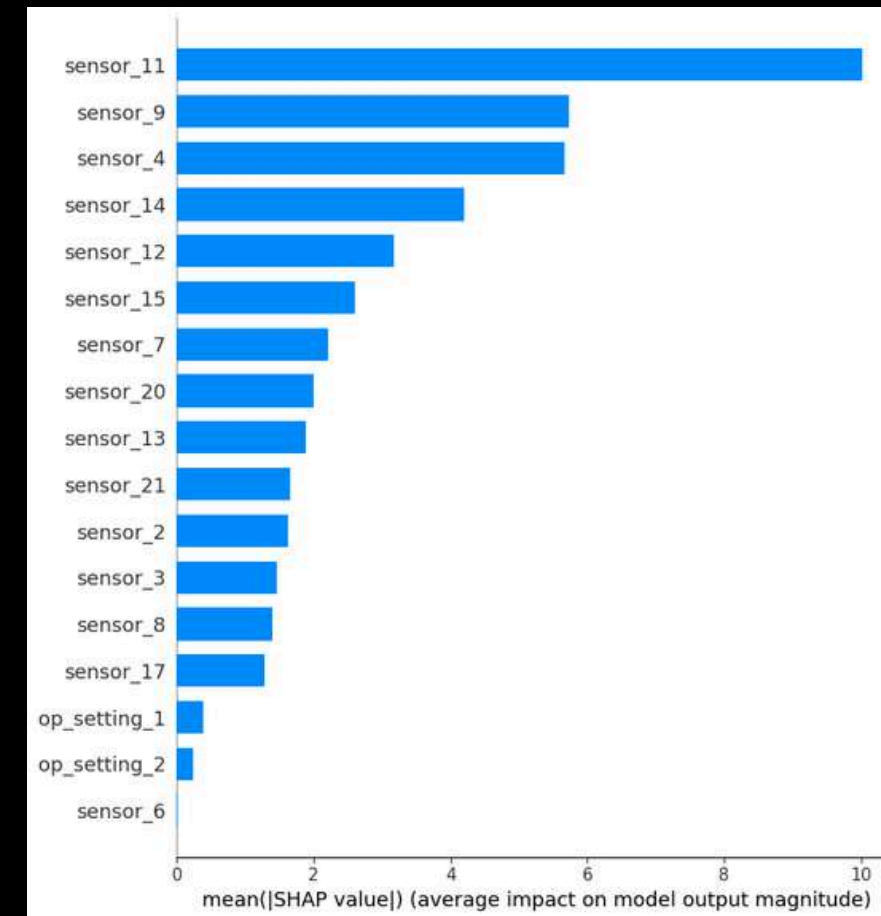
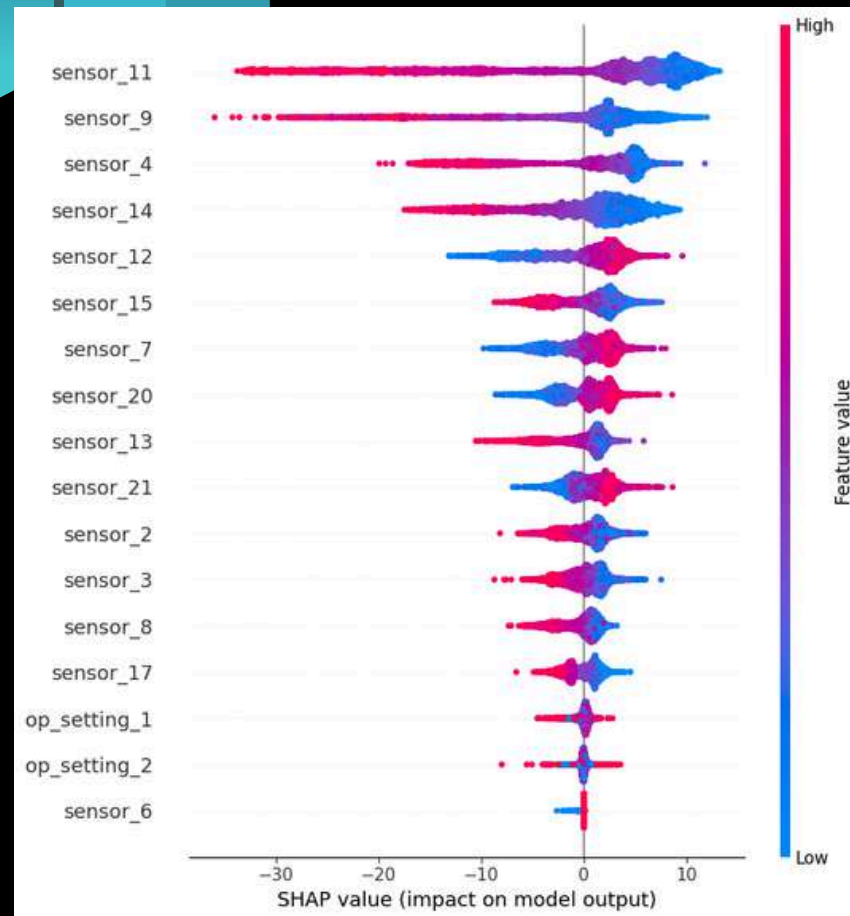
Observation :

- XGBoost achieved the lowest RMSE and MAE.
- It outperformed the LSTM model on the FD001 dataset.

Decision :

- XGBoost was selected as the final production model due to its higher prediction accuracy and better generalization.

EXPLAINABLE AI



Top Features :

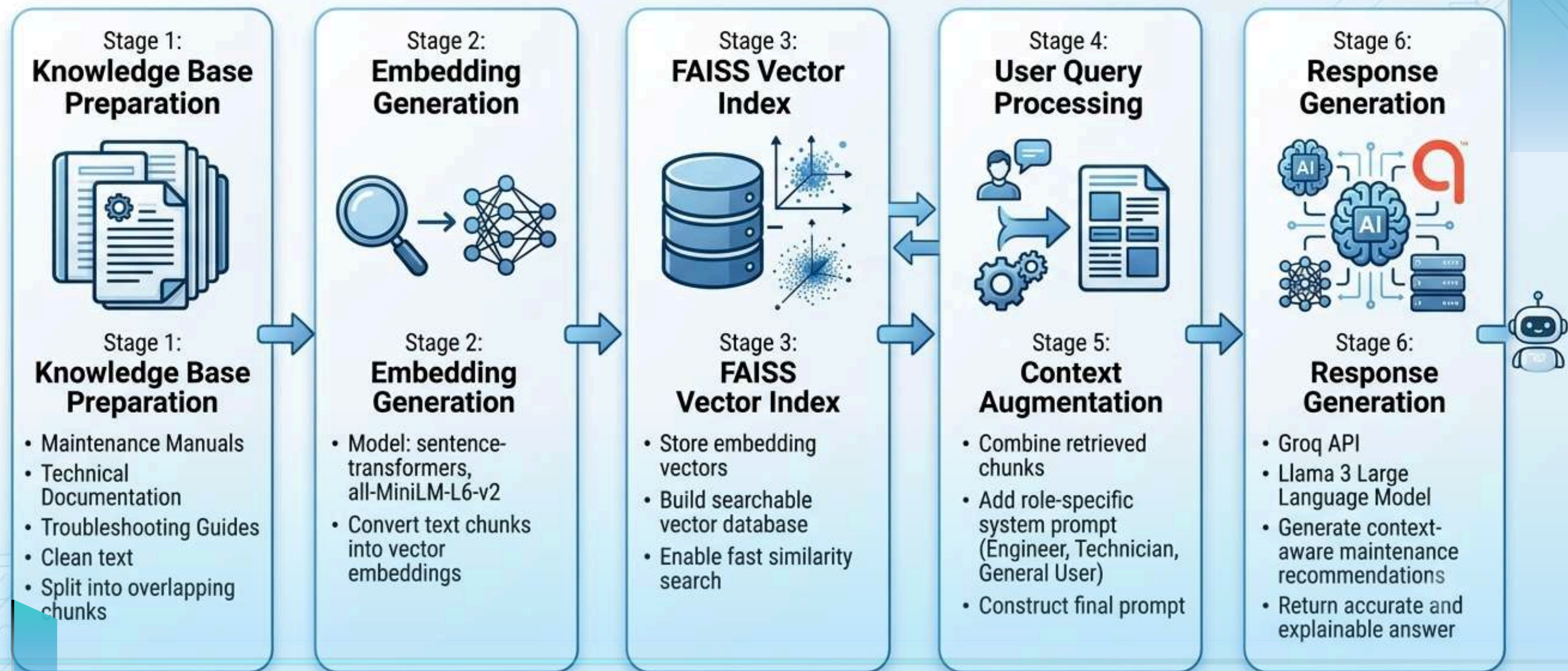
- sensor_11
- sensor_9
- sensor_4
- sensor_12
- sensor_14

SHAP explains how each feature contributes to the predicted Remaining Useful Life.

RAG PIPELINE

Retrieval-Augmented Generation (RAG) Pipeline

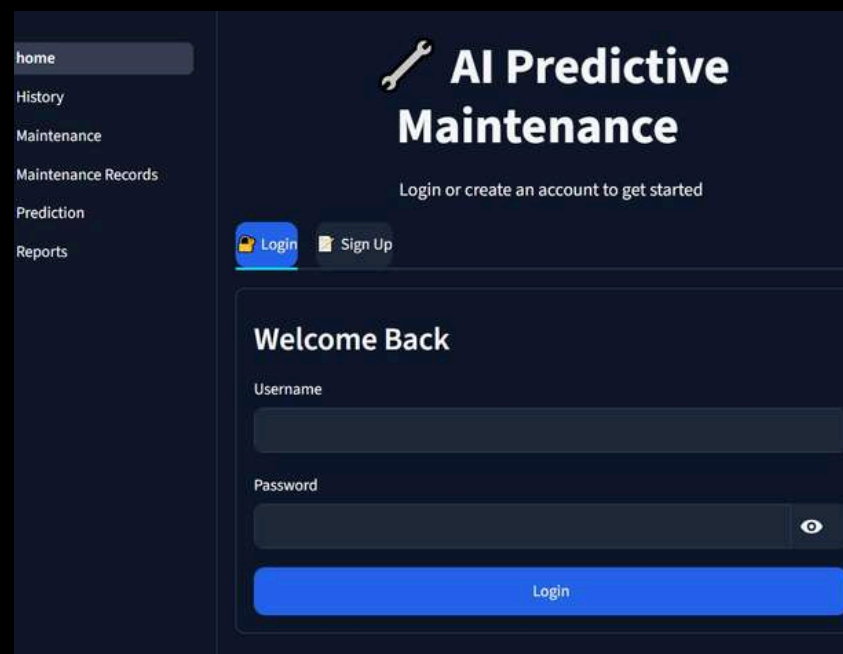
AI Predictive Maintenance System



Embedding Model: all-MiniLM-L6-v2 | Vector Database: FAISS | LLM: Groq-hosted Llama 3 | Retrieval: Top-K Semantic Search

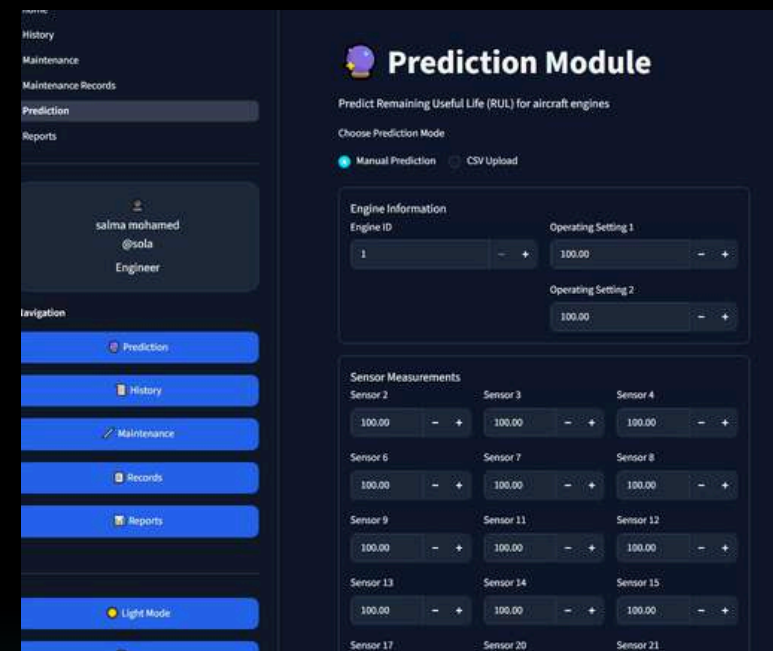
DASHBOARD OVERVIEW

The dashboard is designed to provide engineers with an intuitive interface for monitoring engine health, managing maintenance activities, and analyzing prediction results.



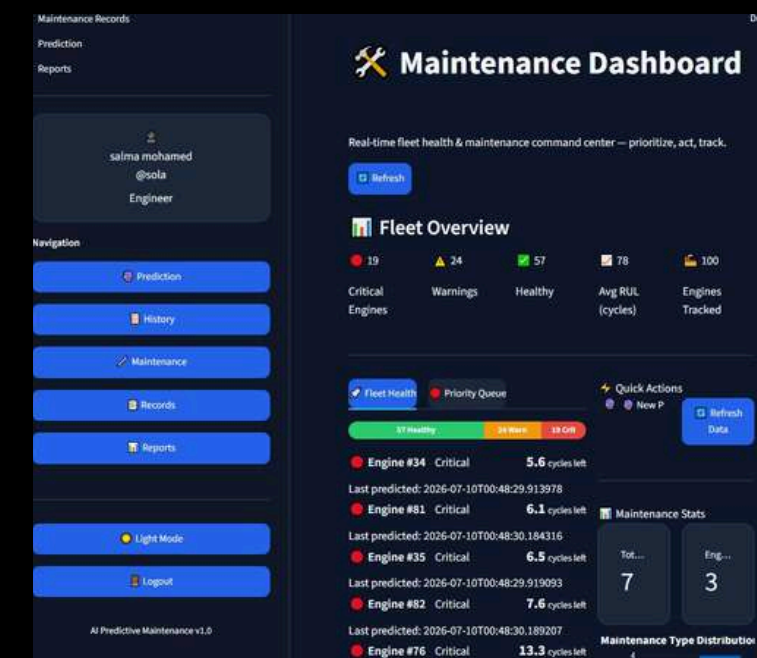
Authentication

- Secure Login & Registration
- Role-based Access



Prediction

- Predict Remaining Useful Life (RUL)
- Display Engine Health Status



Maintenance

- Record Maintenance Operations
- Track Maintenance History



Reports & Analytics

- KPIs Dashboard
- Interactive Charts
- Export Reports (CSV & Excel)

DESIGN 1 – LOGIN & USER AUTHENTICATION

Secure access to the system.

User authentication before using the dashboard.

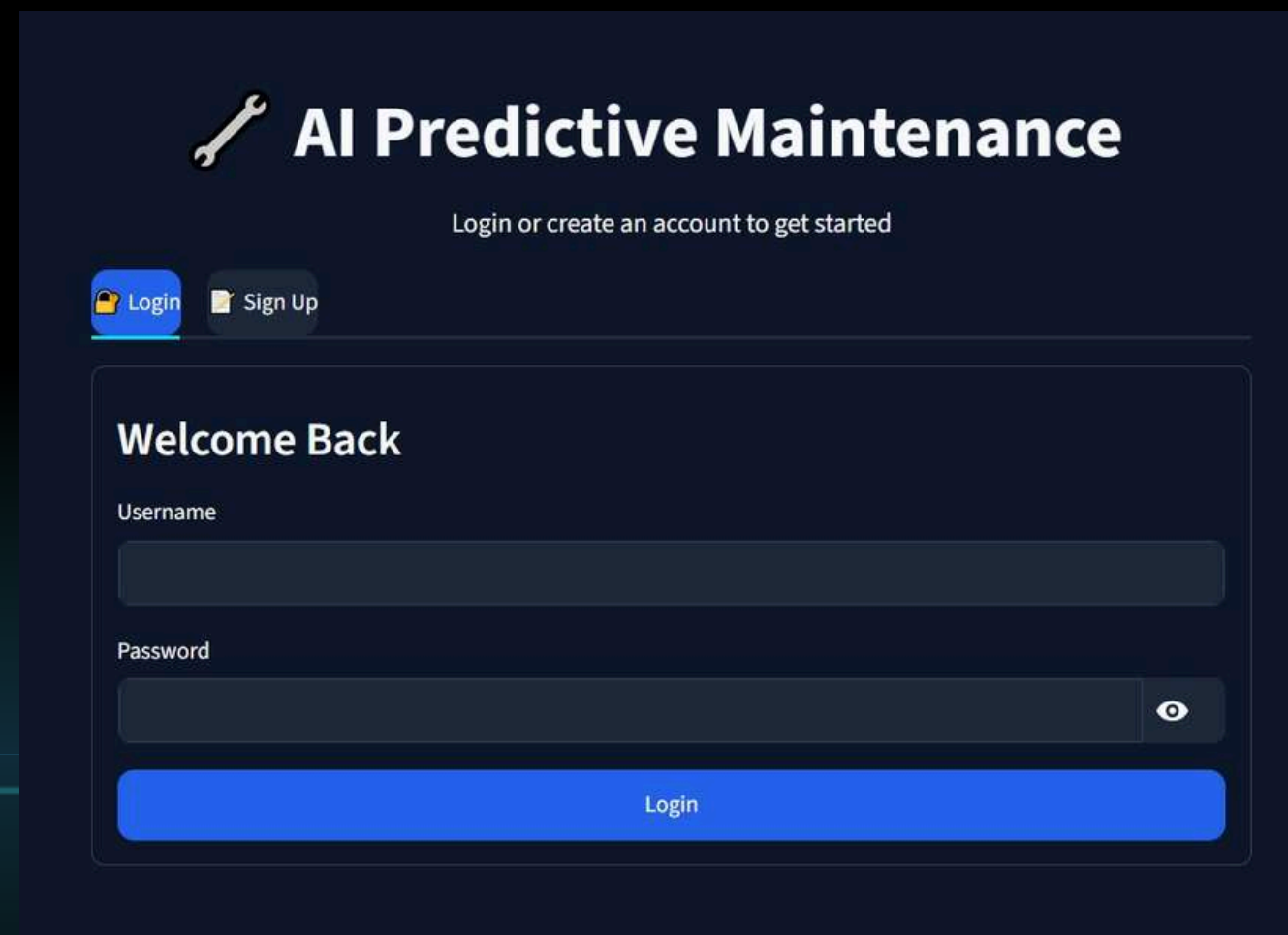
Separate login and registration.

Features

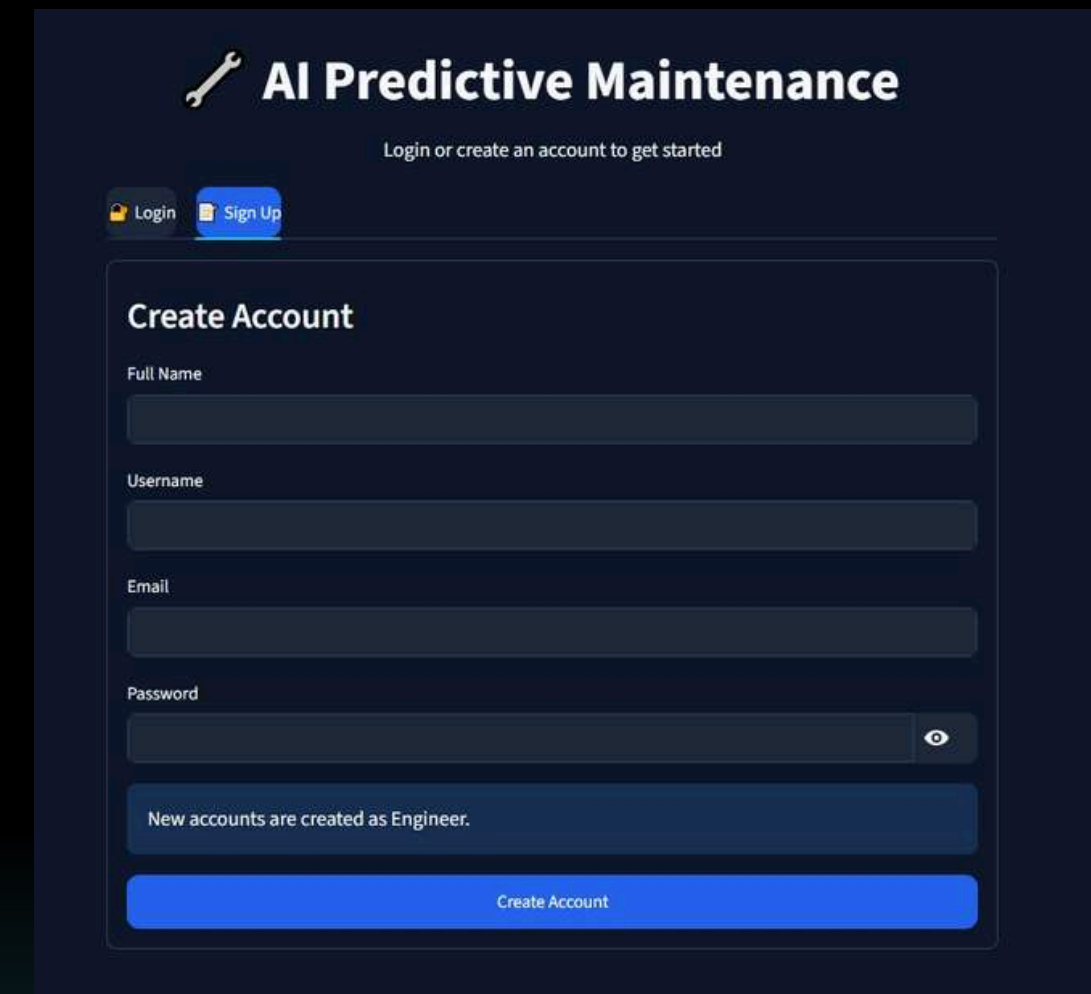
- User Login
- Create New Account
- User Role Management (Engineer / Admin)
- Session Management
- Secure Password Authentication

Technologies Used

- Streamlit (User Interface)
- Python
- SQLite Database
- Authentication Service



The login form for 'AI Predictive Maintenance' features a dark blue header with a wrench icon and the title. Below the title is a subtitle 'Login or create an account to get started' and two buttons: 'Login' (with a lock icon) and 'Sign Up' (with a document icon). The main form area is titled 'Welcome Back' and contains two input fields: 'Username' and 'Password'. The password field has a toggle icon (an eye) to show or hide the password. A large blue 'Login' button is positioned at the bottom of the form.



The 'Create Account' form for 'AI Predictive Maintenance' has a dark blue header with a wrench icon and the title. Below the title is a subtitle 'Login or create an account to get started' and two buttons: 'Login' (with a lock icon) and 'Sign Up' (with a document icon). The main form area is titled 'Create Account' and contains four input fields: 'Full Name', 'Username', 'Email', and 'Password'. The password field has a toggle icon (an eye) to show or hide the password. Below the input fields is a blue button labeled 'Create Account'. A note at the bottom states 'New accounts are created as Engineer.'

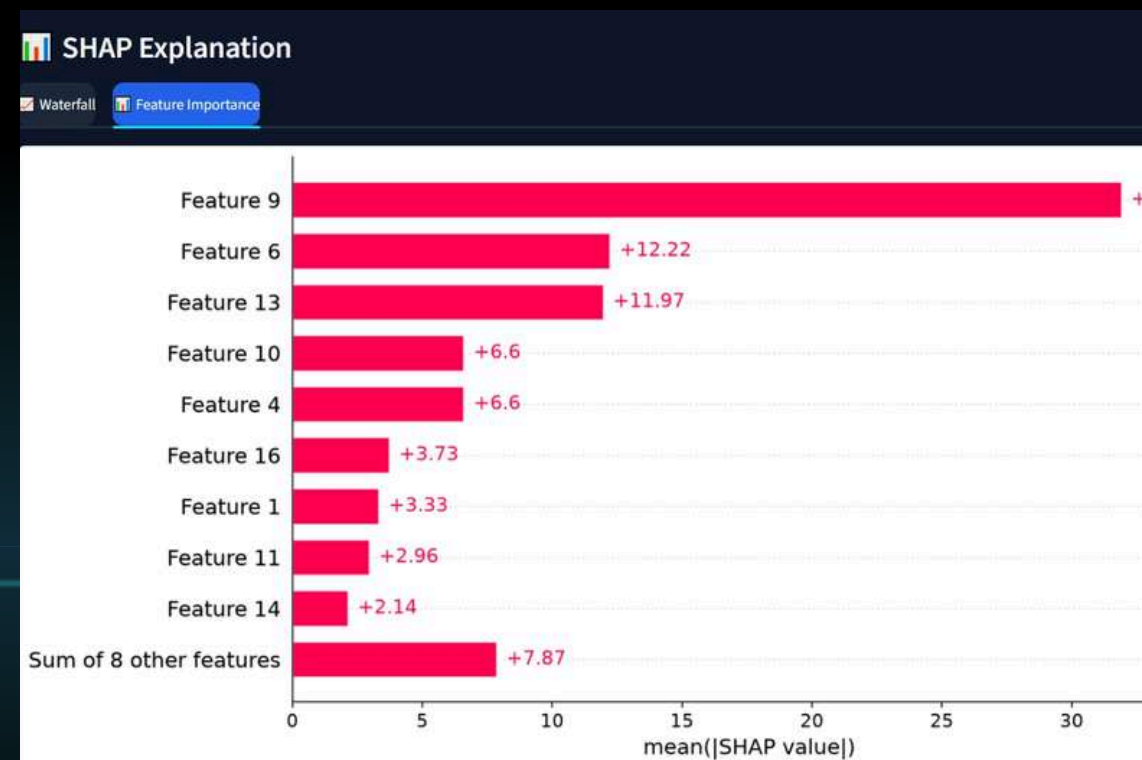
DESIGN 2 – PREDICTION DASHBOARD

Features

- Enter engine sensor values manually or upload engine data.
- Predict the Remaining Useful Life (RUL).
- Display the engine health status (Healthy, Warning, or Critical).
- Save prediction results to the database automatically.
- Explain the prediction using SHAP feature importance.

Technology

- Streamlit
- XGBoost
- SHAP
- SQLite
- Scikit-learn



Prediction Module

Predict Remaining Useful Life (RUL) for aircraft engines

Choose Prediction Mode

☒ Manual Prediction ☐ CSV Upload

Engine Information

Engine ID: 1

Operating Setting 1: 100.00

Operating Setting 2: 100.00

Sensor Measurements

Sensor	Value
Sensor 2	100.00
Sensor 3	100.00
Sensor 6	100.00
Sensor 7	100.00
Sensor 9	100.00
Sensor 11	100.00
Sensor 13	100.00
Sensor 14	100.00
Sensor 17	100.00
Sensor 20	100.00

Predict RUL

Predicted RUL: 31.03 Cycles

Maintenance Recommended Soon

Recommended Action: Schedule maintenance inspection.

DESIGN 3

MAINTENANCE RECORDS

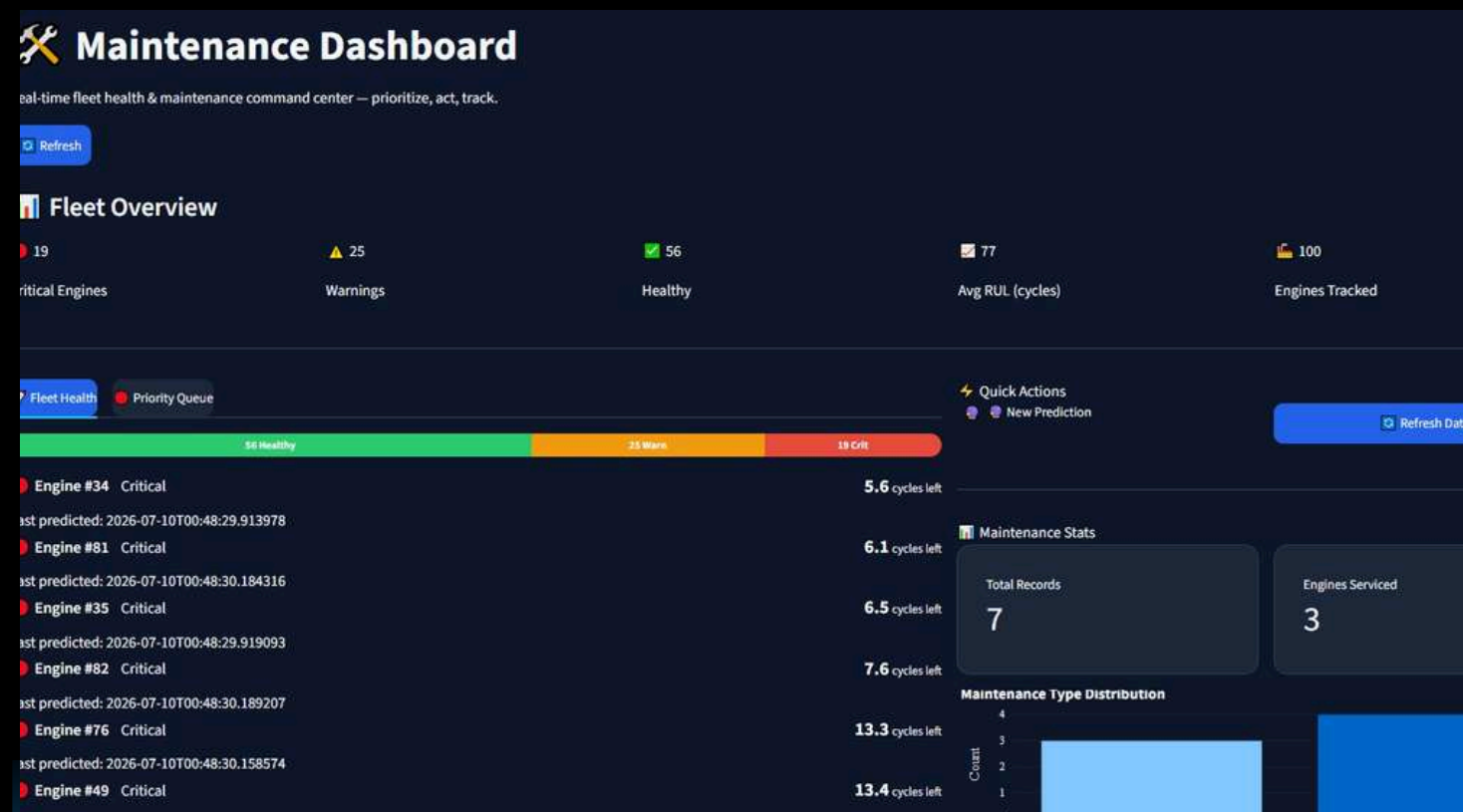
Manage, monitor, and track maintenance activities for each engine while linking them with prediction history.

Main Features

- Log new maintenance records.
- Investigate a specific engine using Engine Drill-Down.
- View all maintenance records in one place.
- Compare maintenance history with prediction history.
- Export maintenance and prediction records as CSV files.

Technology

- Streamlit
- SQLite
- Pandas
- Plotly



Maintenance Records
Log new maintenance, investigate engines, and browse all records.

New Maintenance Record

Engine ID: 1

Maintenance Type: Routine Check

Status: Completed

Cost (\$): 0.00

Downtime (hours): 0.00

Notes (optional): Describe work done, parts used, observations...

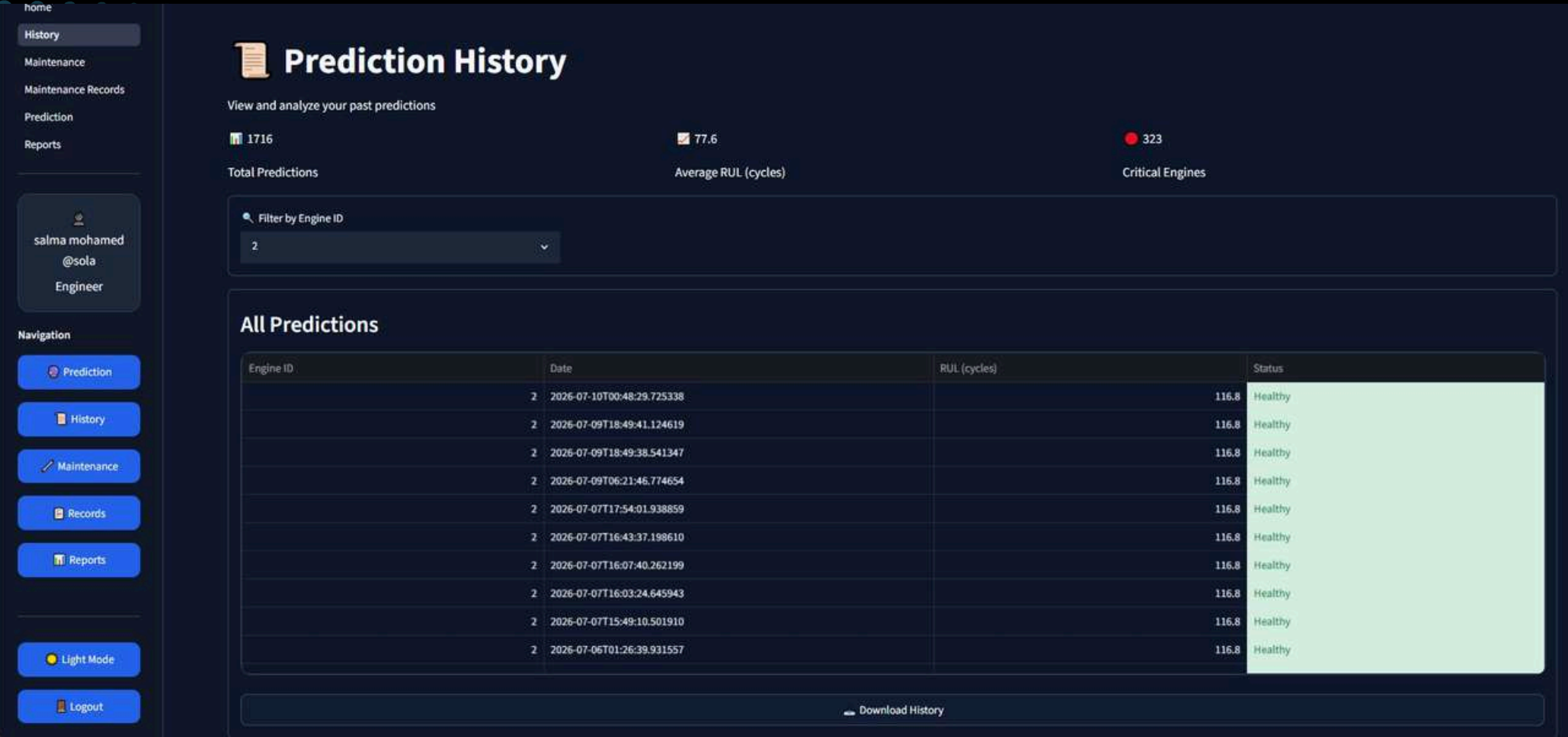
Date: 2026/07/12

About Maintenance Types

Type	When to Use
Routine Check	Scheduled periodic inspection
Repair	Fixing a specific fault or failure
Part Replacement	Swapping a worn/damaged component
Inspection	Detailed examination (may involve teardown)
Overhaul	Major rebuild or comprehensive service
Other	Any maintenance not covered above

DESIGN 4— PREDICTION HISTORY

Store, review, and analyze previous prediction results for each engine.



Features

- View all prediction records.
- Filter predictions by Engine ID.
- Display prediction statistics (Total Predictions, Average RUL, Critical Engines).
- Show engine health status for each prediction.
- Export prediction history as a CSV file.

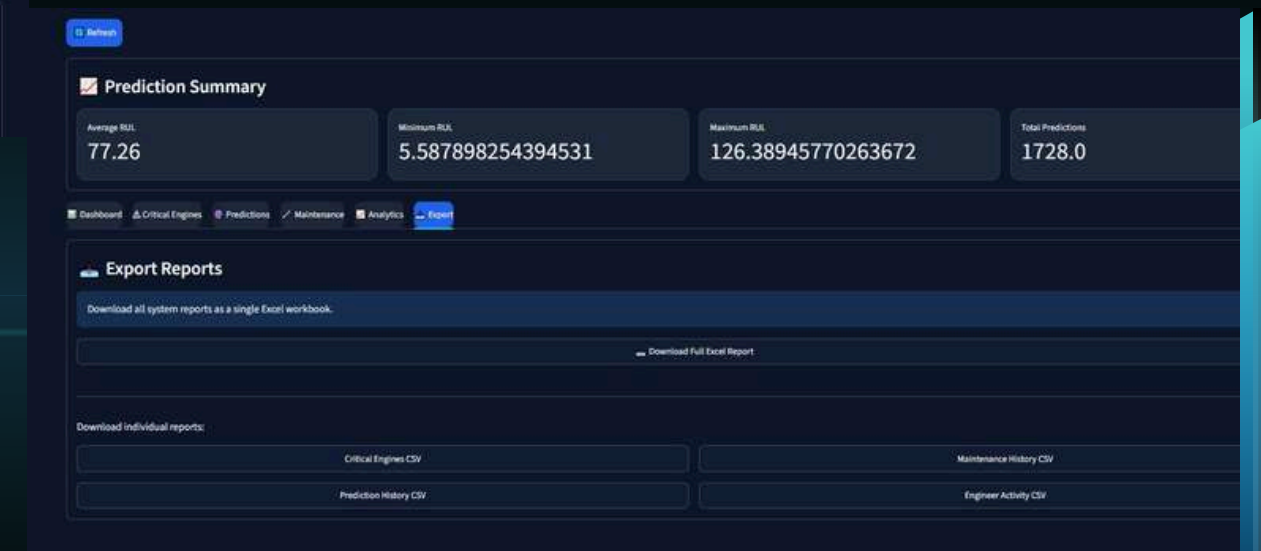
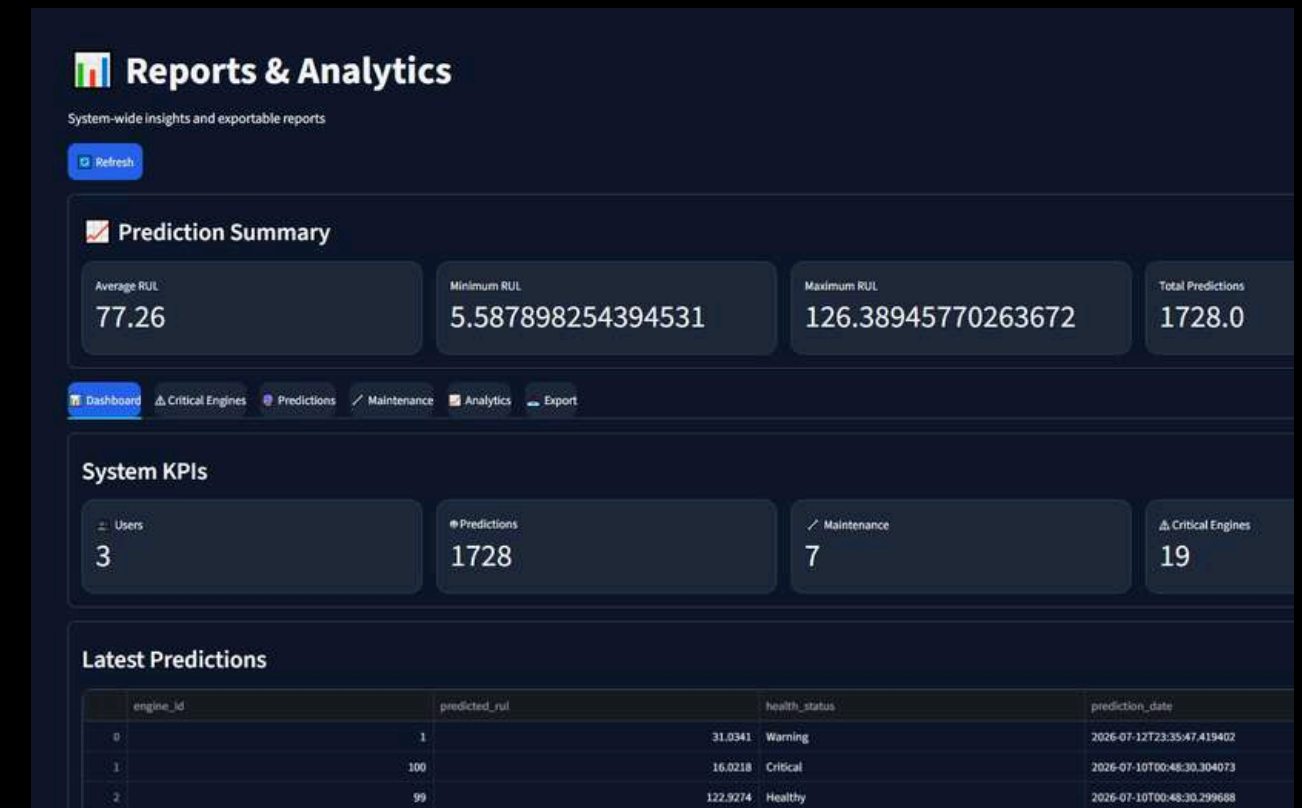
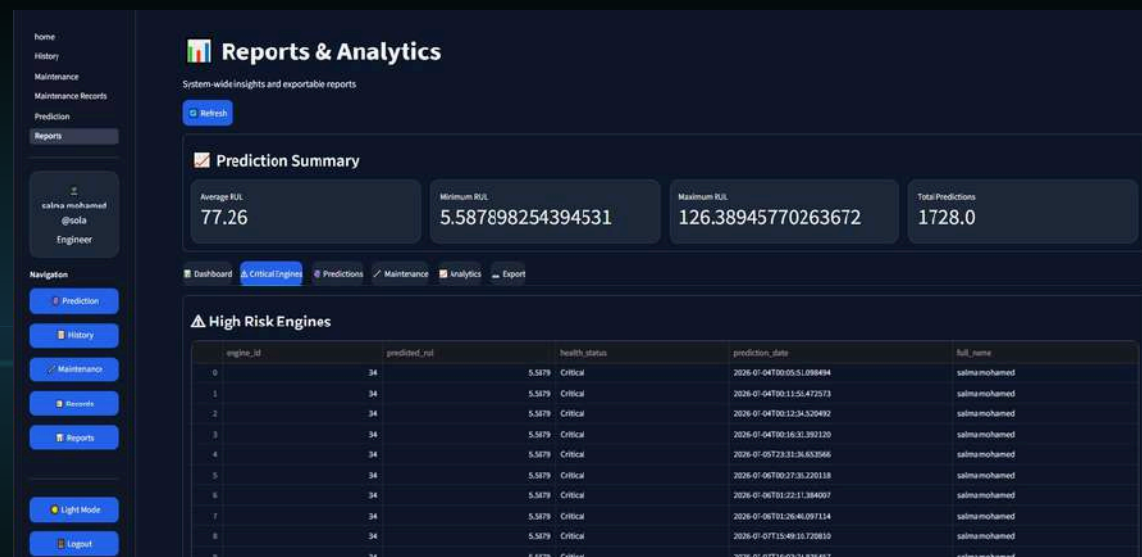
DESIGN 5 – REPORTS

DASHBOARD

Provide interactive reports and visual analytics to support maintenance decisions and monitor engine performance.

Features

- Display key performance indicators (KPIs).
- Visualize RUL prediction trends.
- Show engine health status distribution.
- Analyze maintenance activities.
- Monitor prediction history over time.
- Generate and export reports.



RESULTS



Predictive Maintenance Model

XGBoost model for Remaining Useful Life (RUL) prediction
Evaluated using RMSE, MAE, and R^2 metrics



Explainable AI (XAI)

SHAP identifies influential sensor features
Provides transparent and trustworthy predictions



Intelligent Monitoring Dashboard

Real-time RUL prediction and equipment health monitoring
Maintenance tracking and automated reporting



RAG Maintenance Assistant

Knowledge-based maintenance support
Built using Sentence Transformers, FAISS, and Groq LLM

A complete AI-driven maintenance platform combining prediction, explainability, and intelligent decision support

CHALLENGES & SOLUTIONS

Challenges



1. Sensor Data Complexity

Multivariate time-series sensor data with different operating conditions.



2. Accurate RUL Prediction

Predicting equipment degradation and remaining useful life accurately.



3. Black-Box AI Decisions

Lack of transparency in machine learning predictions.



4. Knowledge-Based Maintenance Support

Providing reliable maintenance recommendations from technical knowledge.



5. System Integration & Usability

Combining AI models, databases, and user interfaces into one system.



6. Model Management

Tracking experiments and managing model versions.

Solutions

1. Data preprocessing, feature selection, and feature engineering pipeline.

2. Machine learning model development and optimization.

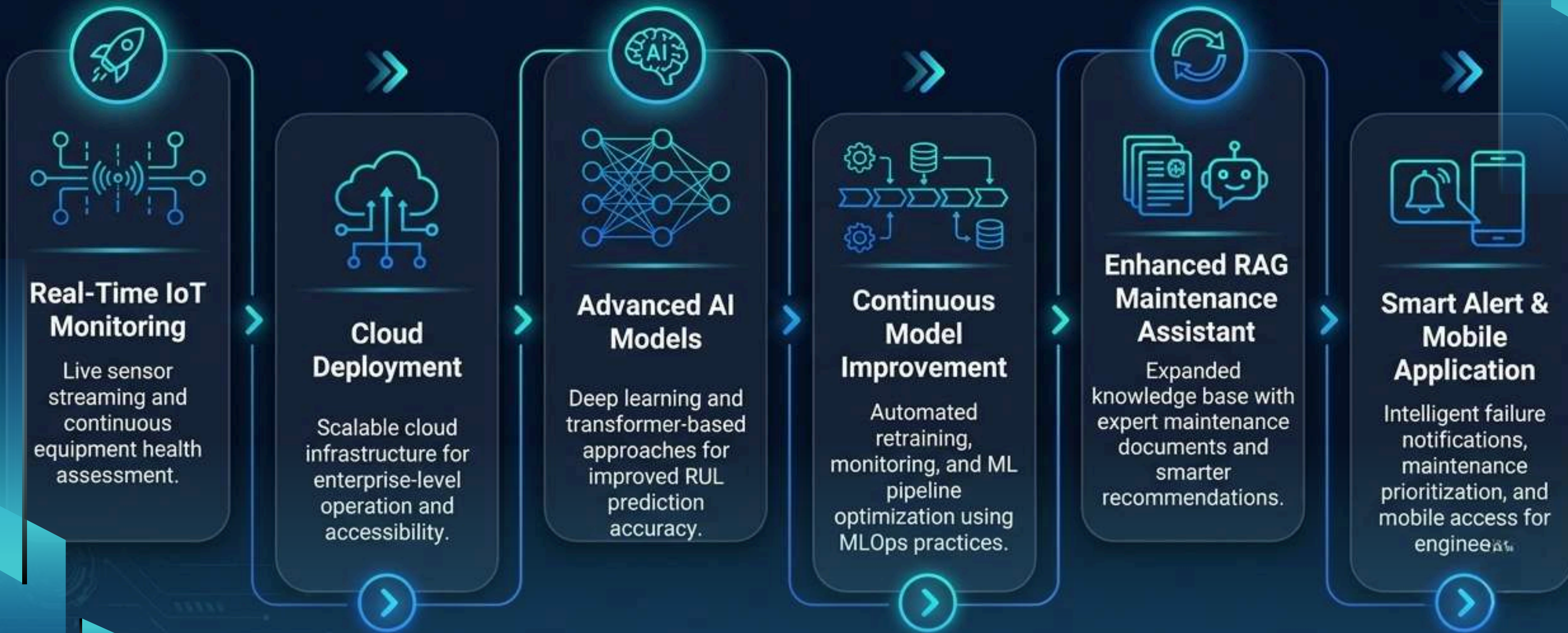
3. SHAP-based Explainable AI integration.

4. RAG pipeline using Sentence Transformers, FAISS, and Groq LLM.

5. Modular architecture with Streamlit dashboard and service layers.

6. MLflow tracking and model lifecycle management.

FUTURE IMPROVEMENTS



CONCLUSION



The background is a dark navy blue. On the left side, there are several overlapping, semi-transparent blue geometric shapes, including rectangles and parallelograms, some of which are tilted. On the right side, there is a grid of small, light blue dots that fades out towards the center.

THANK YOU